



CSE3433 SOFTWARE ARCHITECTURE

PROGRAMME: COMPUTER SCIENCE (SOFTWARE ENGINEERING)

SEMESTER II 2025/2026

GROUP 6

PROJECT TITLE:

SMARTLEARN: CLOUD-NATIVE LEARNING MANAGEMENT SYSTEM (LMS)

INDIVIDUAL LOGBOOK

LECTURER: DR. MOHAMAD NOR HASAN

No.	Group Members	Matric No.	Role
1.	Nurain Nazirah Binti Abdullah	S75415	Backend Developer
2.	Puteri Ireene Hazwani Binti Muhamad Haizar	S75702	Frontend Developer
3.	Nurul Alessa Binti Mohd Nizam	S76278	Full-Stack Architect & Database Lead

Weekly Progress

PHASE 1 (Week 3 – Week 5) – Project Proposal

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 3	Initial system idea & concept formation	Defined SMARTLEARN concept and initial requirement discussion	Unclear system scope and system direction	Conduct deeper research on cloud-native learning management systems and identify how real-world LMS platforms are structured. Break down system into modules such as user management, course management, and system flow. Use AI tools and references to validate concept clarity and ensure project feasibility before proceeding to design phase.
Week 4	Requirement analysis & project validation	Refined LMS idea and explored system features and naming	Difficulty narrowing system scope	Analyse similar LMS platforms and extract only essential features suitable for academic implementation. Separate functional and non-functional requirements clearly. Validate system direction with teammates and refine scope based on lecturer expectations to avoid over-complexity.
Week 5	Proposal drafting & finalisation	Defined functional and non-functional requirements	Time pressure before submission deadline	Review entire proposal structure including objectives, scope, system requirements, and methodology. Ensure all sections are consistent and aligned with

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
				SMARTLEARN system. Perform proofreading, formatting correction, and final validation before submission to ensure completeness and academic compliance.

PHASE 2 (Week 6 – Week 7) – Architecture Design

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 6	Study 3-tier & cloud-native architecture	Researched system architecture and design concepts	Limited understanding of cloud-native architecture	Strengthen conceptual understanding by studying multiple architecture examples and breaking them into presentation layer, application layer, and database layer. Map SMARTLEARN system into 3-tier architecture clearly and identify how cloud deployment fits into each layer.
Week 7	Architecture design and system diagram preparation	Designed component architecture and database structure	Complexity in diagram structuring	Improve diagram structure by ensuring proper relationship between system components. Validate architecture with system requirements and ensure consistency between architecture diagram, database design, and system workflow. Refine based on feedback from team discussion.

PHASE 3 (Week 8 – Week 11) – Prototype Progress

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 8	Prototype development and initial testing	Tested basic system flow and defined Course Management module scope	UI inconsistency in early prototype	Standardise UI layout across all pages to ensure consistent user experience. Conduct usability testing on system flow and refine interface design based on feedback. Improve navigation structure for better user interaction.
Week 9	System evaluation & debugging	Checked database connection and API integration	Database and API integration issues	Debug backend systematically by isolating API endpoints and testing each module individually. Improve backend structure for scalability and maintainability. Ensure proper documentation of API flow for integration consistency.
Week 10	Frontend, backend integration and deployment preparation	Developed login/dashboard and explored deployment setup	First-time deployment experience using Render and Vercel	Follow deployment documentation step-by-step and perform trial deployments. Identify configuration errors such as environment variables and build settings. Refine deployment process until system is stable and accessible online.
Week 11	System enhancement, testing and cloud deployment	Improved UI, fixed bugs and deployed prototype	Deployment configuration and hosting issues	Perform full system testing after deployment to ensure stability. Improve UI responsiveness and fix runtime errors. Monitor cloud deployment performance and continuously troubleshoot issues to ensure system reliability.

PHASE 4 (Week12 – Week 13) - Project Report, Individual Logbook & Final Demo

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 12	Final system review and report preparation	Verified deployed system functionality and reviewed documentation	Documentation structure and content organisation	Validate system against all requirements and testing checklist. Ensure all features function correctly. Follow official report template strictly and reorganise content for clarity, consistency, and academic formatting standards.
Week 13	Final report submission, presentation, and system demo	Completed project report and individual logbook, prepared presentation slides, demo script, and conducted system demonstration	Formatting inconsistencies in documentation and nervousness for presentation	Standardise formatting across all documents, perform final proofreading, conduct multiple rehearsals to improve confidence and presentation flow, and prepare clear technical explanations and answers for the presentation session.

GenAI Usage Record

Week	Date	Tool Used	Purpose	Prompt Used	Output Summary	How It Was Used
Week 3	29/03/2026	ChatGPT	Project understanding	"cuba explain cloud native LMS simple"	Cloud-native concept explained clearly including	Used during initial project planning and idea exploration

Week	Date	Tool Used	Purpose	Prompt Used	Output Summary	How It Was Used
					scalability, flexibility, and cloud deployment characteristics	
Week 3	30/03/2026	ChatGPT	Concept exploration	"cuba bagi contoh system untuk cloud native ni"	Examples of cloud-native systems and LMS platforms provided	Used to compare and evaluate possible project ideas
Week 3	31/03/2026	ChatGPT	Project naming	"Cuba suggest nama untuk project ni"	Generated several project name suggestions	Used to determine SMARTLEARN project identity
Week 4	06/04/2026	ChatGPT	Project validation	"dalam problem statement kena cakap psl apa je"	Explored key elements required in problem statement	Used to finalise project problem statment for proposal
Week 4	07/04/2026	ChatGPT	System analysis	"ape je yang kena ada dalam sistem nii helppps"	Explained system modules, user roles, and workflow requirements	Used to define project scope and system functionalities
Week 5	13/04/2026	ChatGPT	Requirements analysis	"functional requirements LMS apa"	Functional requirements identified including authentication, course management, and enrolment features	Used in proposal preparation
Week 5	14/04/2026	ChatGPT	Requirements analysis	"suggest non functional dalam LMS jap"	Non-functional requirements such as scalability, performance, security, and	Used in software architecture planning

Week	Date	Tool Used	Purpose	Prompt Used	Output Summary	How It Was Used
					availability identified	
Week 5	17/04/2026	ChatGPT	Proposal refinement	"cuba check okay tk proposal ni"	Improved proposal structure, objectives, and project scope description	Used to finalise proposal before submission on 18/04/2026
Week 6	20/04/2026	ChatGPT	Architecture design	"3 tier architecture explain"	Explained presentation layer, business logic layer, and database layer architecture	Used in architecture design documentation
Week 6	21/04/2026	ChatGPT	Cloud architecture design	"eh tapi aku cloud native management system, kena guna cloud mcm mna nk buat"	Suggested cloud deployment architecture and integration approach	Used to justify cloud-native implementation
Week 7	27/04/2026	ChatGPT	System modelling	"SMARTLEARN component diagram"	Suggested system components and UML modelling structure	Used in architecture design preparation
Week 7	28/04/2026	ChatGPT	Architecture refinement	"based on this make sure fullfill this and 3tier arch"	Refined architecture to align with project requirements and three-tier design principles	Used before architecture submission
Week 7	29/04/2026	ChatGPT	Database design	"dalam database table apa je kena ada untuk system ni?"	Proposed database schema including user, course,	Used in database design documentation

Week	Date	Tool Used	Purpose	Prompt Used	Output Summary	How It Was Used
					enrolment, and management tables	
Week 8	11/05/2026	ChatGPT	Module planning	"Dr aku suruh fokus dekat satu module je, kitorang pilih Course Management Module okay tak?"	Validated module selection and implementation scope	Used to define prototype focus
Week 8	12/05/2026	ChatGPT	Scope validation	"flow Course Management Module ni okay tak?"	Refined system workflow and module interaction flow	Used before prototype development
Week 9	18/05/2026	ChatGPT	Development planning	"basic coding structure LMS"	Suggested project folder structure and coding organisation	Used during development setup
Week 9	19/05/2026	ChatGPT	API design	"API endpoints login system"	Proposed authentication and system API endpoints	Used in backend development planning
Week 9	20/05/2026	ChatGPT	System improvement	"apa lagi boleh improve?"	Suggested user interface and backend improvements	Used to enhance system quality
Week 10	25/05/2026	ChatGPT	Deployment guidance	"cara deploy backend render"	Provided backend deployment steps using Render	Used during backend deployment

Week	Date	Tool Used	Purpose	Prompt Used	Output Summary	How It Was Used
Week 10	26/05/2026	ChatGPT	Frontend deployment	"frontend deploy vercel step"	Explained frontend deployment using Vercel	Used during frontend deployment
Week 10	27/05/2026	ChatGPT	Debugging support	"error deploy kenapa jadi macam ni"	Diagnosed deployment issues and suggested solutions	Used for troubleshooting deployment errors
Week 11	01/06/2026	ChatGPT	UI enhancement	"UI nampak plain apa nk tambah eh"	Suggested visual and usability improvements	Used to improve user interface design
Week 11	02/06/2026	ChatGPT	Dashboard design	"dashboard LMS design idea"	Recommended dashboard layout and interface structure	Used for frontend enhancement
Week 11	03/06/2026	ChatGPT	Login page enhancement	"login page design better macam mana"	Suggested improvements for login interface design	Used to enhance user experience
Week 12	08/06/2026	ChatGPT	System testing	"check system flow betul tak"	Validated workflow consistency and feature integration	Used during final testing phase
Week 12	09/06/2026	ChatGPT	Bug analysis	"possible bug dalam LMS apa"	Identified potential system bugs and weaknesses	Used for debugging and system refinement
Week 12	10/06/2026	ChatGPT	QA checklist	"final testing checklist apa perlu"	Generated comprehensive testing checklist	Used for quality assurance validation

Week	Date	Tool Used	Purpose	Prompt Used	Output Summary	How It Was Used
Week 13	15/06/2026	ChatGPT	Documentation support	"logbook tulis mcm mna"	Provided logbook writing guidance and structure	Used for project documentation preparation
Week 13	16/06/2026	ChatGPT	Report compilation	"combine observation table"	Organised and combined report tables into a consistent format	Used for final report preparation
Week 13	17/06/2026	ChatGPT	Final project review	"check final report and system completeness"	Reviewed project completeness, documentation quality, and system readiness	Used before final submission
Week 13	18/06/2026	ChatGPT	Submission preparation	"final submission checklist apa perlu"	Generated final submission checklist and verification guide	Used to ensure all project deliverables were complete before submission
Week 13	19/06/2026	ChatGPT	Final presentation preparation	"apa je point penting yg kena letak dlm slide presentation, pstu apa possibility soalan yg Dr akan tanya "	Suggested presentation content, project demonstration flow, key discussion points, and possible questions related to system architecture, implementation, and deployment.	Used to finalise presentation slides, organise the demo session, and prepare responses for potential questions during the final project presentation.